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1. Introduction

Technology is evolving every day and we can see its integral to the application of lean manufacturing in the supply chain industry where it's applied to 115 companies practicing lean manufacturing (Kamble S. et.al., 2020). According to the research conducted by Buer S. (2018), lean manufacturing has helped in increasing productivity. A research article by Touriki F.E. (2021) discusses the need for lean manufacturing to improve productivity while simultaneously maximizing productivity.

Lean manufacturing is a management philosophy that aims to eliminate waste and improve operational efficiency by value-adding activities and continuous improvement (Shah et al., 2002; Womack et al., 2010).

Lean manufacturing can help in eliminating waste, reducing lead times, and improving overall efficiency and quality (Womack, 2010). As per the research article from Mohanty R. et.al. (2023), lean manufacturing can also be successfully used in the automobile industry.

1.1. Background of the Study

Based on the report from Pakistan Automotive Manufacturer's Association (PAMA), the automobile industry has had a significance impact on Pakistan's GDP. The PAMA report from 2021 suggests hit contributes 2.8% to the GDP which amounts to Rs.30 billion. This figure may vary upon the inclusion of other factors such as auto parts manufacturing, auto repair and maintenance services, and retail sales of vehicles. The UNIDO (2021) report highlights the significant impact of auto parts and car retail sales on the economy. According to CEIC (2023), the automobile sector sales growth is recorded at 5.6%. The growth in Pakistan's automobile industry has led to more employment (Agyemang M., 2019).

Moreover, the automotive industry has a significant impact on other sectors of the economy, such as steel, rubber, glass, and plastics. The industry is a major buyer of raw materials and components from these sectors, thereby creating more jobs and contributing to their growth. Thus, the automotive sector has the potential to further boost economic growth in the country. Furthermore, research by Osama (2018) shows that there is an expected increase in automobile manufacturing.

Studies have shown that lean manufacturing can lead to improved productivity, quality, and customer satisfaction (Shah et al., 2002; Shah and Ward, 2007). For example, a study by Taj (2008) found that the implementation of lean manufacturing principles, such as just-in-time production and continuous improvement, resulted in significant improvements in manufacturing performance for a Chinese company.

Several frameworks and tools have been developed to support the implementation of lean manufacturing principles (Shah and Ward, 2007). One such framework is the Lean Transformation Framework, which was developed by the Lean Enterprise Institute to guide organizations in their lean journey (Womack and Jones, 2010). The framework emphasizes the importance of creating a culture of continuous improvement, identifying value streams, and implementing lean tools and techniques.

Value stream mapping has been used to optimize the automobile industry in the past. According to the research conducted by Pradeep A, (2022), value stream mapping was used to reduce lead time in the rubber component's manufacturing process.

1.2. Automobile Manufacturing

1.2.1 Bottlenecks:

When workloads are too large or too fast for the production line to handle, a bottleneck can become an issue. Inefficiencies caused by this backup can lead to delays and higher production costs. The bottleneck refers to the shape and size of the bottle. It is also the place where the bottle is most likely to become congested. This slows down the liquid flow.

Delays in manufacturing can lead to a dramatic increase in production costs and time. When a company starts the production process. There are many ways to organize your lean six sigma processes of a new product, they are more vulnerable to bottlenecks. Because there could be problems in the production process that the company needs to fix, this situation will require more scrutiny and fine-tuning. Operations management focuses on controlling production, identifying bottleneck effects early, and finding effective solutions.

Production Variations

A variance is a difference in production results between the budgeted and actual results.

Managers examine variances to adjust and eventually eliminate bottlenecks. Managers may find that labor costs are higher than expected and decide to make changes. Labor costs can be decreased if management can eliminate the bottleneck.

If materials are left on the factory floor, waiting to be used, a bottleneck could also lead to material variance. You can solve bottlenecks by increasing capacity utilization and finding new suppliers.

Production Capacity

A bottleneck can affect the production capability that a company can achieve each month.

Theoretical capacity is the ability of a company to always produce at its maximum capacity.

This assumes that there are no machine breakdowns, breaks in the bathroom, or vacations for employees.

Businesses use their practical capacity to manage production because theoretical capacity is unrealistic. This capacity includes downtime for machine repairs as well as employee time off. Practical capacity is the range of processes that can be operated efficiently without crashing. If you go beyond the optimal range, the possibility of a bottleneck caused by a breakdown in one or more processes increases. There are many options available to a company if its production capacity is not sufficient to meet its production goals. Management could choose to reduce their production goals to make them more in line with their production capabilities. They could also work together to find ways to simultaneously reduce production bottlenecks and increase production. To meet their production goals, companies often use capacity requirement planning tools and methods. .

Short-term and long-term bottleneck effects

Short-term

Short-term bottlenecks can be caused by an operator's absence or delay in receiving materials from another vendor because of one-off shipping issues.

Long-term

These are problems that recur and have a significant impact on manufacturing processes. Some examples include:

- There is no real-time data to identify patterns within the production
- An inadequate material tracking system results in a lack of traceability
- Delay in production due to audit procedures and paper-based work instructions

While short-term bottlenecks can be isolated and do not require regular attention, long-term bottlenecks will need to be addressed.

Bottleneck Analysis:

The bottleneck will be the one with the longest queue. These red flags can be used to determine if a target step is causing a blockage in the overall workflow.

Accumulation

An accumulation occurs when the input exceeds what a machine can handle. Because the work order can't be processed at the same speed as other steps in manufacturing, inventory and work hours can build up.

Full capacity

The bottleneck in the production process will be a unit or machine that is using most of its full potential. This is easily compared with the lower capacity utilization rates of other units on the shop floor. The bottleneck unit will stop working if the input is increased.

Slow communication

Is there any miscommunication? Or slow communication that could lead to slower manufacturing processes? Low physical products can be caused by poor communication at all levels.

Throughput and throughput times

Each machine's throughput can be increased individually to determine which one has the greatest impact on overall production output. A longer-than-average throughput could also lead to delays in inspection and move times, as well as an increase in the waiting time. It may be necessary to investigate further to determine which step is taking too long.

Steps to detect a workflow Bottleneck:

. If your workflow is chaotic and unpredictable, it's likely that you don't have smooth-flowing operations. It is important to pinpoint the problem and set a countermeasure. You can use several analysis tools to identify bottleneck effects in LEAN Management.

In just three steps, you can identify a bottleneck.

Visualize. Tracking your work makes it easy to see which work items are getting piled up. This is usually a sign that there is a problem or bottleneck.

Map Queues & Activities. By separating activities from queues and mapping them we can see how long our work is sitting in a queue before a specific activity. You have identified your bottleneck if the queue grows faster than the activity stages processes.

Measure the Cycle Time Per Stage. By measuring the cycle time at each stage, you can create a heat map diagram of the cycle time. These workflow stages could also be called queues. .

Optimization of Bottleneck Effects:

Following are the steps to optimize Bottleneck Effect:

Work in batches

Some operations will take less time if similar work items are organized in batches. Be cautious. Higher risk is associated with larger batches. A smaller batch is better. However, in real life, sometimes we must compromise.

Increase the number of people and resources

To speed up the process, you should increase the capacity at the bottleneck. Keep your eyes open. Once the system's resources have been re-distributed, another bottleneck will appear.

Set WIP limits

Consider setting a WIP limit if it does not exist.

Do not leave the bottleneck idle

The ripple effect of flow should be considered. Always load the bottleneck at its maximum capacity.

Work to Reduce the bottleneck

Ensure that the work is delivered in the best possible form. If the review process becomes a bottleneck, make sure that quality is maintained from the beginning. You must ensure that the work being reviewed is flawless. Every error the reviewer discovers is going to cost more time and money.

Bottleneck Analysis tools**Theory of constraints (TOC)**

A method to find the constraint, test it and improve it.

Identify the problem: Look again for signs or speak with the person who raised it. People closest to the problem will be able to provide information and offer solutions.

Use the constraint: Do not let it stop. This will make it easier to see the effects and keep them consistent. This step should not take too long.

All other steps must be subordinated to the constraint. Let the Takt time of the constraint determine the rest. You will need to observe what else must be changed to match the Takt time, and then evaluate the implications for the rest.

Lift the constraint: In this instance, “elevate” means to improve the process around the constraint or bottleneck. To reduce mistakes, do this as fast as possible.

Continue the TOC process. Making changes at one stage of the process can cause a bottleneck at the next. Be prepared to continue this process until you have improved the entire line. Clear communication channels between the project team and the other operators can help streamline the process of identifying and implementing improvements. This can help everyone to see the effects of any bottlenecks or improvements. Communication is the key to reducing frustrations and spreading rumors.

Lean Six Sigma DMAIC Process

This five-step methodology uses data to improve a process.

1. Define the project.
2. Measure process activities, effects, and ability to meet established specifications.

3. Use the data you have collected to identify the root cause. (Helpful tools are Root Cause Analysis and Failure Mode Analysis and Effects or a Multivariable chart).
4. Find ways to eliminate the root cause by improving the process. (Helpful tools are Design for Experiments and Kaizen Events)
5. Control and monitor performance using tools such as 5S and Poke-yokes..

1.2.2 Work In Process Inventory:

Work in process (WIP) inventory refers to materials that are waiting to be assembled and sold. WIP inventory includes the cost of raw materials, labor, and overhead costs needed to manufacture a finished product. Since WIP inventory takes up space and can't be sold for a profit, it's generally a best practice for product-based businesses to minimize the amount of WIP inventory they have on hand. Work in process inventory encompasses all inventory types in the intermediate stage between raw materials inventory and final products. If raw material is combined with direct labor but is not ready to be sold, it counts as WIP inventory.

Work in process vs. work in progress inventory

These two terms are interchangeable, for the most part. However, some supply chain managers might use work in process in the context of production operations that can be completed in a short period of time (like our coffee example above) and use work in progress for large-scale productions like construction projects.

WIP inventory falls in the production process, look at it in the larger context of other inventory classifications. All of the following terms are under the umbrellas of manufacturing inventory.

1. Raw materials inventory: Anything needed to manufacture a product
2. Work in process inventory: All materials waiting to be assembled
3. Finished goods inventory: Assembled materials ready to be sold

Now that you've got a grip on what WIP inventory is, you might be wondering why it's important to classify in the first place.

WIP inventory eats up space in a storage space or factory floor that could otherwise be used for inventory that's ready to sell, thus increasing carrying costs. The more WIP you have on hand, the more capital you have tied up in items waiting to be sold. Too much WIP inventory increases the risk of materials becoming lost, broken, expired, or obsolete before they have a chance to be assembled.

VALUE STREAM MAPPING Value Stream Mapping (VSM) tool shows all the activities, from supply to final product through the different processing steps. In other words it is a sketch of a production line, Value stream map is the techniques that bring the all processing steps at one place. It shows the big picture of shop floor rather than individual processes and improving the each area at the production line. It is used to draw attention to different wastes and eliminating them in future state map. It is necessary that every process as closely as possible produces only what its customers need when they need it.

Value Stream Mapping (VSM) is used to define and analyze the current state for a product value stream and design a future state focused on reducing waste, improving lead-time, in-process inventory and improving workflow. The production flow was optimized thus minimizing several non-value added activities/times such as bottleneck time, waiting time, material handling time, etc.

Implementation of lean tools in any manufacturing unit. Their results show a significant improvement in productivity, reduction of Production Lead Time and reduction in inventory. These can be achieved by creating flow by layout modification and balance to TAKT time,
$$\text{TAKT Time} = \text{Net available time per day} / \text{Customer demand per day}.$$

1.3. Research Gaps

There is a variety of research done on value stream mapping and its applications in the automobile industry. However, there is a research gap identified in value stream mapping being used to identify bottle necks in the muffler assembly and optimizing the process for in process inventory. Furthermore, there is no significant study on the muffler assembly specifically in Karachi, Pakistan.

1.4. Problem Statement

Loads Limited faces high costs as a result of long lead times and inventory levels. Improving the low production efficiency will enable the company to save costs and potentially provide customer satisfaction.

1.5. Research Objectives:

The purpose of our research is to cover the following objectives:

- Study and analyze the current state of Loads Limited's assembly line in Pakistan.
- To identify the bottlenecks and their impact on the manufacturing process.
- Share recommendations and suggestions to optimize the in process inventory using value stream mapping

1.6. Research Questions

The research aims to answer the research questions mentioned below:

Q1. What are the bottlenecks?

Q2. How much time will value stream mapping save the company?

Q3. Can it be applied to any other area to optimize the process?

1.7. Scope of the Study

Using value stream mapping, we can identify bottlenecks, reduce wastage, and optimize the in process inventory. It will help us capture the current view of the process and the time consumed by each stage of the process.